

Felipe A. Cordera Urrutia

PhD Candidate, MIT Operations Research Center

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Research Interests: Optimization for large-scale operational systems, balancing model fidelity and computational scalability, with particular interest in complex downstream effects and uncertain or unavailable information.

Education

Massachusetts Institute of Technology Cambridge, MA
PhD in Operations Research Expected May 2027

Universidad de Chile Santiago, Chile
Master of Science in Engineering — Water Resources and Environment 2020
Passed with highest distinction. Thesis: “Stochastic Dual Dynamic Programming Applied to Reservoir Operations: Limitations in Uncertainty Modeling and Presentation of an Efficient Method to Overcome Them.”

Universidad de Chile Santiago, Chile
Professional Degree in Civil Engineering — Water Resources and Environment 2020
Passed with highest distinction; ranked first in the graduating class.

Universidad de Chile Santiago, Chile
Bachelor of Science in Engineering — Civil Engineering 2016
Passed with highest distinction.

Research Experience

MIT Operations Research Center Cambridge, MA
Research Assistant 2022–present
Supervisors: Alexandre Jacquillat and Cynthia Barnhart

- **Robotic fulfillment:** Develop large-scale optimization methods for assigning customer orders and inventory pods to workstations, accounting for downstream robot movements through travel estimates and simulation.
- **Advanced air mobility:** Develop scalable optimization methods that jointly coordinate eVTOL vehicle dispatch and trajectories through capacity-constrained air-corridor networks.
- **Airport operations:** Develop tractable representations of connecting-passenger impacts for large-scale gate assignment, accounting for walking distances and tight connection times.

Amazon Robotics North Reading, MA
Research Scientist II Intern May–Aug. 2025
Supervisor: Michael Caldara

- Developed a congestion-aware integer optimization model for large-scale warehouse robotic fulfillment, assigning customer orders and inventory pods to pick stations using predicted congestion-dependent travel distances.
- Extended a high-fidelity discrete-event simulator of robotic traffic with real-time conflict resolution, adding inventory and order-fulfillment dynamics.
- Evaluated the proposed optimization methods on full-scale fulfillment-center instances, demonstrating improvements in robot travel, congestion, and workstation utilization.

Universidad de Chile Santiago, Chile
Research Assistant — Resilient Planning of Low-Carbon Power Systems 2020–2022
Supervisors: Rodrigo Moreno and Fernando Ordóñez

- Developed multistage stochastic optimization methods for power-system operations under correlated renewable-generation uncertainty, including a variant of Stochastic Dual Dynamic Programming (SDDP) that explicitly captures temporal correlations.
- Applied the methodology to unit commitment with energy storage on benchmark systems and the Chilean power system, evaluating the value of adaptive operational decisions under uncertainty.

Institute for Complex Engineering Systems, Universidad de Chile
Project Engineer — Power-System Reliability and Transmission Planning
Supervisor: Rodrigo Moreno

Santiago, Chile
2020–2022

- Developed an inverse optimization framework to infer component-level outage and repair parameters from observed system-level reliability metrics and integrate them into probabilistic transmission planning.
- Applied the framework to the large-scale Chilean transmission system to quantify the costs and benefits of alternative reliability levels, informing the development of Chile’s national transmission reliability standard.

Cornell University, Civil and Environmental Engineering
Visiting Student, Prof. Patrick Reed’s Research Group

Ithaca, NY
May–Jul. 2018

- Developed and evaluated multi-objective reservoir operating policies using evolutionary optimization and simulation, as part of master’s thesis research.

Publications and Working Papers

1. C. Barnhart, M. Caldara, **F. Cordera**, O. Islam, A. Jacquillat, and A. Schmid. “Order Picking in Large-scale Robotic Mobile Fulfillment Systems.” *Working paper; targeted for submission to Production and Operations Management*.
2. **F. Cordera**, A. Jacquillat, and S. McDonald. “Optimizing Advanced Air Mobility Operations in a Corridor Network.” *INFORMS Optimization Society Conference Refereed Proceedings, 2026; submitted to INFORMS Journal on Optimization*.
3. F. Sepúlveda, D. Alvarado, **F. Cordera**, E. Esperguel, G. Strbac, and R. Moreno. “Cost-Benefit Analysis of Transmission Network Reliability Standards: An Inverse Optimal Power Flow Approach.” *IEEE Access*, 2026.
4. **F. Cordera**, R. Moreno, and F. Ordóñez. “Unit Commitment Problem with Energy Storage Under Correlated Renewables Uncertainty.” *Operations Research*, 71(6), 1960–1977, 2023.
5. S. E. Null, M. A. Olivares, **F. Cordera**, and J. R. Lund. “Pareto Optimality and Compromise for Environmental Water Management.” *Water Resources Research*, 57(10), e2020WR028296, 2021.

Research Presentations

- “**Order Picking in Large-scale Robotic Mobile Fulfillment Systems.**” MIT Center for Transportation and Logistics (CTL) Research Seminar, Massachusetts Institute of Technology, Cambridge, MA, 2026. *Invited talk*.
- “**Optimizing Advanced Air Mobility Operations in a Corridor Network.**” INFORMS Transportation Science and Logistics Society Conference, Massachusetts Institute of Technology, Cambridge, MA, 2026.
- “**Optimizing Advanced Air Mobility Operations in a Corridor Network.**” INFORMS Optimization Society Conference, Atlanta, GA, 2026.
- “**Order Picking in Large-scale Robotic Mobile Fulfillment Systems.**” INFORMS Annual Meeting, Atlanta, GA, 2025.
- “**Short-Term Operation Scheduling of Renewable Power Systems under Multistage Uncertainty.**” LXXXIX Annual Meeting of the Chilean Mathematical Society (SOMACHI), Universidad de O’Higgins, Rancagua, Chile, 2021.

Teaching Experience

Massachusetts Institute of Technology
Teaching Assistant, 15.083 Integer Optimization

Cambridge, MA
Spring 2025

Massachusetts Institute of Technology
Teaching Assistant, 15.093 Optimization Methods

Cambridge, MA
Fall 2023

Universidad de Chile
Teaching Assistant, EL4114 Optimization

Santiago, Chile
Fall 2022

Universidad de Chile
Teaching Assistant, CI71A Water Resources Systems Analysis

Santiago, Chile
2017–2020

Universidad de Chile
Teaching Assistant, CI5103 Environmental Systems Analysis

Santiago, Chile
Fall 2017

Academic Service

- **Co-organizer**, MIT Operations Research Center IAP Seminar, “*GenAI Meets OR*,” Massachusetts Institute of Technology, January 2026.
- **Conference Volunteer**, INFORMS Transportation Science and Logistics Society Conference, Massachusetts Institute of Technology, July 2026.

Selected Applied Projects

- **Medical Appointment and Patient Transportation Scheduling**. Developed optimization methods for the Chilean Safety Association — Institute for Complex Engineering Systems, Universidad de Chile, 2019–2020.
- **Medical Degree Revalidation Scheduling**. Developed an integer optimization tool for examination construction and simulated-patient assignment — Faculty of Medicine, Universidad de Chile, 2020.
- **Preference-Based Assignment of Medical Interns**. Developed a deferred-acceptance-based tool for assigning interns to clinical rotations — Faculty of Medicine, Universidad de Chile, 2019–2020.
- **University Course Scheduling**. Developed an integer optimization model for interdisciplinary course scheduling — Faculty of Medicine, Universidad de Chile, 2018–2019.

Honors and Awards

- **ENRE Student Best Paper Award**, “Unit Commitment Problem with Energy Storage Under Correlated Renewables Uncertainty,” INFORMS Annual Meeting, 2024.
- **Best Graduate Engineer of the 2020 Class**, Chilean Association of Engineers, 2021.
- **National Master’s Scholarship**, National Council for Science and Technology (CONICYT, now ANID), Chile, 2017.
- **Short-Term Research Program Award**, Universidad de Chile, 2017. Funding for a research visit to Cornell University.
- **Pablo Pérez Zañartu Scholarship**, San Carlos de Maipo Foundation, 2016. Awarded to the top incoming student in Civil Engineering, Water Resources and Environment, at Universidad de Chile.
- **Dean’s List for Outstanding Academic Performance**, Universidad de Chile, 2011–2013 and 2015.

Technical Skills

Methods: Large-scale optimization (integer, network, stochastic), decomposition methods, machine learning, discrete-event simulation

Programming: Julia, Python, C, MATLAB; Linux

Software: Gurobi, FICO Xpress, GAMS

Languages: Spanish (native), English (high proficiency)